TINKERFORGE SENSORY RIG — CIRCUIT OVERVIEW

Water Reaching Task | ESP32 Brick + 5 Bricklets | 12 V DC Industrial Supply

POWER STAGE

12 V / 2 A ADAPTER
Bel SW124-12-E-P5
+12 V Output | **GND** Common

ESP32 POWER SUPPLY
SKU 42 | Input: 7–27 V
Output: Regulated 5 V (ESP32)

POWER DISTRIBUTION RAILS
- **+12 V:** Relay load bus rail
- **5 V:** ESP32 Brick operating voltage
- **GND:** Common system reference

CONTROLLER

ESP32 BRICK
SKU 113 | Firmware 2.3.1
Port A • Port B • Port C • Port D
USB-C: Host PC Configuration / Data

HOST PC INTERFACE
USB-C Data / Telemetry & Setup

PORTS E & F (SPARE)
IO-4 V2 / SKU 2124 | Camera Trigger

BRICKLETS & CONNECTED DEVICES

PORT A: DI 4 2.0
Opto-isolated CH0 → BNC Camera Sync (Adafruit 2889)

PORT B: TOUCH 2.0
MPR121 (E0) → Metal Spout (Lick Detection Sensor)

PORT C: PIEZO 2.0
SKU 2145 | Built-in Speaker (50 Hz–15 kHz Acoustic Cue)

PORT D: QUAD RELAY 2.1
Solid-State | SKU 2102

30 V / 600 mA | CH0: Spare

QUAD RELAY OUTPUTS & LOADS (POLARITY-SENSITIVE DUAL TERMINAL PINOUT)

Solid-State Switching: Strict Polarity Required

CH1 — BLUE LED (1092C6-12V)

+ CH1 (+) | − CH1 (−)

+12 V **Red Wire:** +12 V Rail → Relay **CH1 (+)** input terminal.

LOAD **Green Wire:** Relay **CH1 (−)** output → Blue LED Anode (+).

GND **Black Wire:** Blue LED Cathode (−) → Common **GND** return rail.

Built-in series resistor for direct 12 V DC operation.

CH2 — SOLENOID VALVE (075P2NC12)

+ CH2 (+) | − CH2 (−)

+12 V **Red Wire:** +12 V Rail → Relay **CH2 (+)** input terminal.

LOAD **Green Wire:** Relay **CH2 (−)** output → Solenoid coil (+) lead.

GND **Black Wire:** Solenoid coil return lead → Common **GND** rail.

• 1N4007 Flyback Diode across coil (cathode / silver stripe to green switched line).

CH3 — GREEN LED (1092C6-12V)

+ CH3 (+) | − CH3 (−)

+12 V **Red Wire:** +12 V Rail → Relay **CH3 (+)** input terminal.

LOAD **Green Wire:** Relay **CH3 (−)** output → Green LED Anode (+).

GND **Black Wire:** Green LED Cathode (−) → Common **GND** return rail.

Built-in series resistor for direct 12 V DC operation.

Physical Assembly Warning: Because Quad Relay 2.1 utilizes solid-state switching, the **(+)** and **(−)** terminals are polarity-dependent. The Red wire must be connected to the relay (+) terminal, and the switched Green load line to the (−) terminal. Reverse connection will cause failure to switch or permanent hardware damage.

WIRING & CONNECTION LEGEND:

- **Red: +12 V Supply → Relay (+)**
- **Green: Switched Output Relay (−) → Load**
- **Black: Load Return → System GND**
- **Blue: Bricklet Signal / 5 V Bus**